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Roll no.

Degree: B.Tech, Semester: Third-Course work
END-SEMESTER EXAMINATION, Nov-Dec, 2023

Course Code: ECMTC04

Course Title: Mathematics for Signal Processing and Communication

Duration: 03 Hours

Max. Marks: 50

Note: Attempt all the five questions. Missing data/ information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
Q1	Attempt any 2 parts of the following:		
1a	Define abelian group. Show that if a, b are any two elements of a group $(G, *)$, then $(ab)^2 = a^2b^2$ if and only if G is abelian.	5	CO1
1b	Prove that the intersection of two subgroups of a group G is a subgroup of G .	5	CO1
1c	Define ring homomorphism. Let z be a complex no. so that $z = a + ib$ for some $a, b \in R$, and $\bar{z} = a - ib$ be its complex conjugate. Prove that $\phi: C \rightarrow C$, defined as $\phi(z) = \bar{z} \quad \forall z \in C,$ is a ring homomorphism with respect to addition and multiplication.	5	CO1
Q2	Attempt any 2 parts of the following:		
2a	If the vectors $(0, 1, x)$, $(x, 1, 0)$, $(1, x, 1)$ of the vector space $R^3(R)$ are linearly dependent, then find the value of x .	5	CO2
2b	If $\{u_1, u_2, u_3\}$ is a basis of $R^3(R)$, then show that $\{u_1 + u_2, u_2 + u_3, u_3 + u_1\}$ is also a basis of $R^3(R)$.	5	CO2
2c	State Rank-Nullity theorem. Let $T: R^3(R) \rightarrow R^3(R)$ is a linear transformation defined as $T(a, b, c) = (a - b + 2c, 2a + b - c, -a - 2b) \quad \forall (a, b, c) \in R^3.$ Find the rank and nullity of T .	5	CO2
Q3	Attempt any 2 parts of the following:		
3a	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = \vec{r} $, then show that $\text{div}(\text{grad } r^m) = m(m+1)r^{m-2}.$	5	CO3

3b	Evaluate the surface integral $\iint_S (yz\hat{i} + zx\hat{j} + xy\hat{k}) \cdot \vec{ds},$ where S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$ in the first octant.	5	CO3
3c	Use Green's theorem to evaluate $\int_C (2x^2 - y^2)dx + (x^2 + y^2)dy,$ where C is the boundary of the area enclosed by the x-axis and the upper half of the circle $x^2 + y^2 = a^2$.	5	CO3
Q4	Attempt any 2 parts of the following:		
4a	Determine the analytic function $f(z) = u + iv$, if $v = \ln(x^2 + y^2) + x - 2y$.	5	CO4
4b	Evaluate the following integral using Cauchy's integral formula: (i) $\oint_C \frac{3z+5}{z^2+2z} dz$, C is the circle $z = 1$. (ii) $\oint_C \frac{e^z}{(z+1)^2} dz$, C is the circle $z - 1 = 3$.	5	CO4
4c	State Residue Theorem. Determine the poles of the function $f(z) = \frac{z^2}{(z-1)(z-2)^2}$ and the residue at each pole.	5	CO4
Q5	Attempt any 2 parts of the following:		
5a	A string is stretched between the fixed points $(0,0)$ and $(l,0)$ and released at rest from position $y = y_0 \sin \frac{\pi x}{l}$. Find the displacement $y(x,t)$.	5	CO5
5b	Solve heat equation $\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}$, given that $u(0,t) = u(l,t) = 0$ and $u(x,0) = x$, where l is the length of the bar.	5	CO5
5c	Solve the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ satisfying the following conditions: $u(0,y) = u(L,y) = u(x,0) = 0$ and $u(x,a) = \sin\left(\frac{n\pi x}{L}\right)$.	5	CO5